LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 7 Test (Set B): How do Organisms Reproduce?

Chapter / Topic: How do Organisms Reproduce?

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SECTION A

Q1. Multiple fission is seen in: **[1]**

Answer: (c) Plasmodium

Q2. Which organism reproduces by fragmentation? **[1]**

Answer: (d) Spirogyra

Q3. Which part of a flower develops into the fruit? **[1]**

Answer: (a) Ovary

Q4. The transfer of pollen grains from the anther to the stigma is called: **[1]**

Answer: (d) pollination

Q5. Which method of contraception also helps protect against sexually transmitted diseases? **[1]**

Answer: (c) Condom

Q6. **Assertion (A):** A unisexual flower contains either stamens or a pistil, not both. **Reason (R):** Papaya and watermelon have unisexual flowers. **[1]**

Answer: (b). R gives examples; it does not explain A.

SECTION B

Q7. Give two advantages of sexual reproduction over asexual reproduction. **[2]**

Answer: (1) It combines DNA from two parents, so it creates more variation. (2) Variation increases the chance that some individuals survive if the environment changes, which helps the species survive.

- Each advantage: 1

OR, answer: Bisexual flowers have both stamens and a pistil (e.g. hibiscus, mustard). Unisexual flowers have either stamens or a pistil (e.g. papaya, watermelon).

- Each with example: 1

Q8. Why does the lining of the uterus become thick and spongy every month? **[2]**

Answer: Every month the uterus prepares to receive and nourish a fertilised egg. The lining becomes thick and spongy, with a rich blood supply, so that the embryo can implant and be nourished.

- To receive fertilised egg: 1
- Rich blood supply to nourish embryo: 1

SECTION C

Q9. Answer the following about puberty: **[3]**

(i) Boys: testosterone. Girls: oestrogen. [1]

(ii) (Any two) Thick hair grows in the armpits and genital area; the skin becomes oily and pimples may appear; height and weight increase quickly. [1]

(iii) Testes (in boys) and ovaries (in girls). [1]

Q10. What is germination? Name the parts of a seed that take part in it, and give the role of each. **[3]**

Answer: Germination is the development of the embryo inside a seed into a seedling under suitable conditions. Plumule: the future shoot. Radicle: the future root. Cotyledon: stores food for the growing seedling. The seed coat protects the seed.

- Definition: 1
- Parts with roles: 2

OR, answer: Each species occupies a niche, and reproduction replaces individuals that die, keeping the population steady. Because DNA is copied fairly consistently, offspring keep a body design suited to that niche. Small variations from DNA copying mean that if the niche changes, some individuals may still survive, so the species continues.

- Replaces dying individuals: 1
- Consistent DNA copying keeps body design: 1
- Variation helps survival in a changed niche: 1

SECTION D

Q11. Answer the following: **[5]**

(i) DNA copying is never absolutely accurate, so small changes appear in the copies, and offspring differ slightly from parents and from each other. These variations may help some individuals survive if the environment changes, for example a rise in temperature. [2]

- DNA copying errors cause variation: 1
- Usefulness: 1

(ii) Cutting: a piece of stem with buds is planted in moist soil and grows roots (rose, sugarcane). Layering: a branch is bent down and covered with soil while still attached to the parent; once it grows roots it is cut off (jasmine). Grafting: the stem of one plant (scion) is joined onto the rooted stem of another (stock) so that they grow as one plant (mango, rose). [3]

- Each method: 1

OR, answer: (a) Diseases spread through sexual contact. Bacterial: gonorrhoea, syphilis. Viral: warts, HIV-AIDS. (b) Using a condom (barrier method). (c) The thick, spongy lining of the uterus is no longer needed. It breaks down and comes out through the vagina as blood and mucus. This is menstruation, which happens roughly every month and lasts about 2–8 days.

- (a) 2
- (b) 1
- (c) lining breaks down 1, menstruation 1

SECTION E

Q12. When a Planaria is cut into several pieces, each piece grows into a complete organism. Specialised cells in the body divide to make large numbers of… **[4]**

(i) The ability of a fully formed organism to grow new individuals from its body parts. [1]

(ii) Hydra [1]

(iii) Complex animals have highly specialised cells, tissues and organs. Most of their cells cannot divide to form all the cell types needed for a whole new body. Regeneration is not the same as reproduction, because most organisms do not normally depend on being cut up in order to reproduce. [2]

- Complexity/specialised cells: 1
- Not the same as reproduction: 1